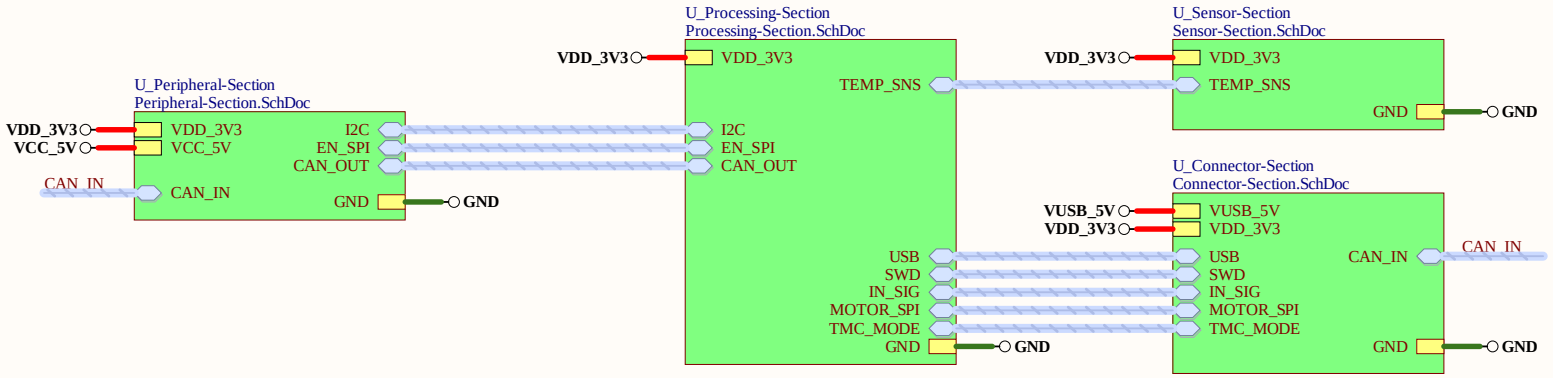
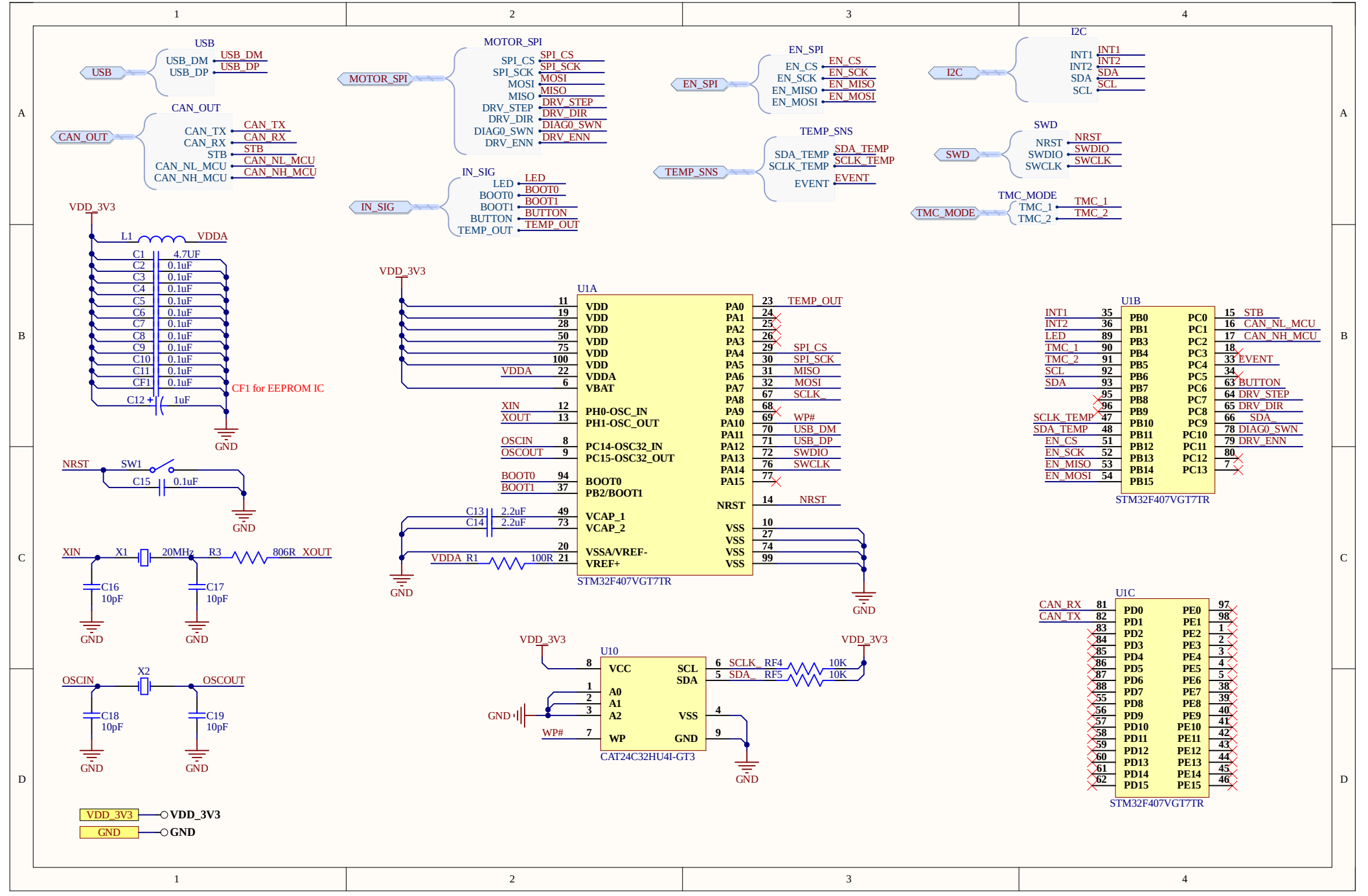
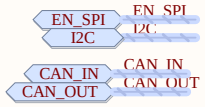
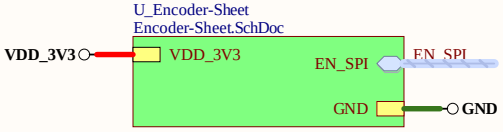
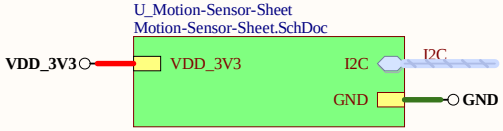
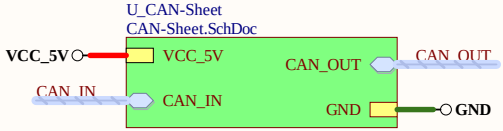

U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
I2C
EN_SPI
CAN_OUT
GND
GND
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
I2C
EN_SPI
CAN_OUT
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
GND
GND
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
U_Connector-Section
Connector-Section.SchDoc
VUSB_5V
VDD_3V3
VUSB_5V
VDD_3V3
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
GND
CAN_IN
CAN_IN
GND
GND

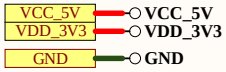

USB
USB_DM
USB_DP
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
IN_SIG
LED
BOOT0
BOOT1
BUTTON
TEMP_OUT
EN_SPI
EN_CS
EN_SCK
EN_MISO
EN_MOSI
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
I2C
INT1
INT2
SDA
SCL
SWD
NRST
SWDIO
SWCLK
TMC_MODE
TMC_1
TMC_2
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
CF1 for EEPROM IC
C12 1uF
GND
NRST
SW1
C15 0.1uF
XIN
X1
20MHz
R3
806R
XOUT
C16 10pF
C17 10pF
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
U1A
VDD
VDDA
VBAT
PH0-OSC_IN
PH1-OSC_OUT
PC14-OSC32_IN
PC15-OSC32_OUT
BOOT0
PB2/BOOT1
VCAP_1
VCAP_2
VSSA/VREF-
VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NRST
VSS
STM32F407VGT7TR
C13 2.2uF
C14 2.2uF
VDDA_R1 100R
U1B
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
U1C
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
U10
VCC
SCL
SDA
A0
A1
A2
VSS
WP
GND
CAT24C32HU4I-GT3
SCLK_ RF4 10K
SDA_ RF5 10K
WP#
VDD_3V3
GND

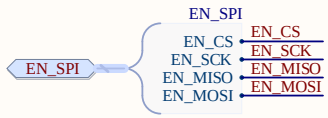

EN_SPI
EN_SPI
I2C
I2C
CAN_IN
CAN_IN
CAN_OUT
CAN_OUT

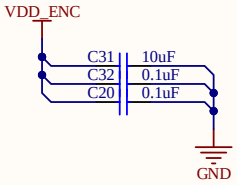

U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND
GND

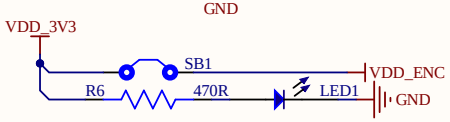

U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND
GND

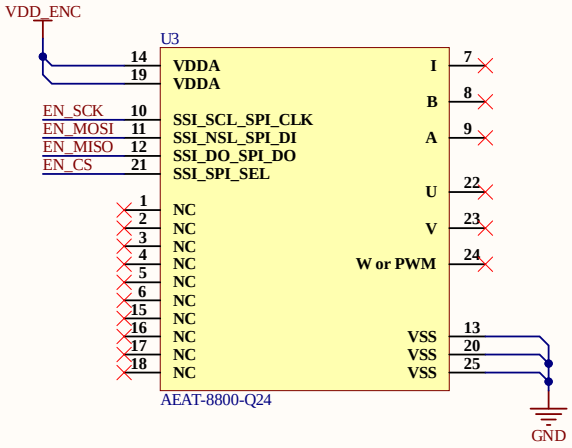

U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
CAN_IN
GND
GND
GND

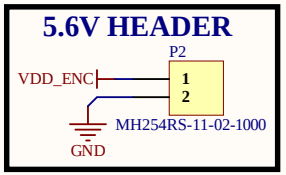

VCC_5V
VCC_5V
VDD_3V3
VDD_3V3
GND
GND



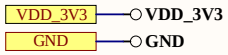


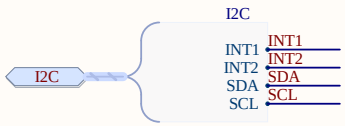


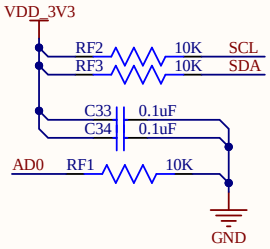


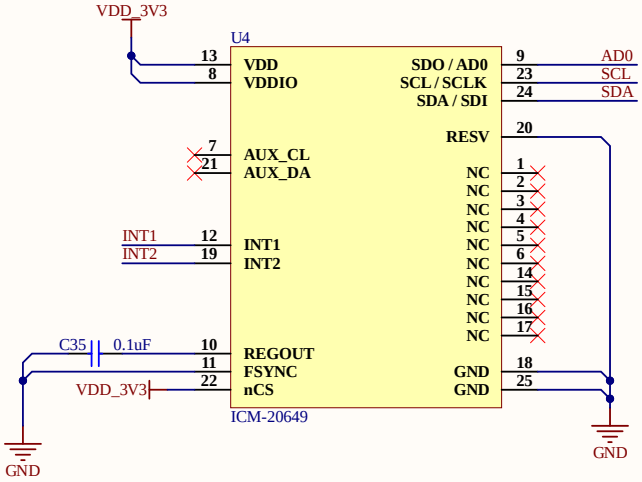


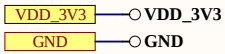
This header will be used to provide external power (5.6V) to Encoder in order to program its OTP Registers. For this, SB1 bridge needs to be disconnected.

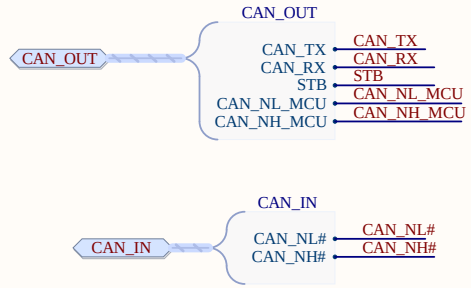


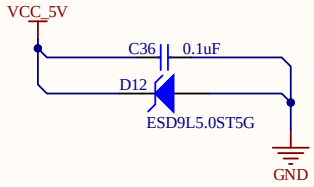

I2C
I2C
INT1
INT2
SDA
SCL
INT1
INT2
SDA
SCL


VDD_3V3
RF2
10K
SCL
RF3
10K
SDA
C33
0.1uF
C34
0.1uF
AD0
RF1
10K
GND


VDD_3V3
U4
13
VDD
8
VDDIO
SDO / AD0
SCL / SCLK
SDA / SDI
9
AD0
23
SCL
24
SDA
RESV
20
7
AUX_CL
21
AUX_DA
NC
1
2
3
4
5
6
14
15
16
17
INT1
12
INT2
19
INT1
INT2
C35
0.1uF
10
11
22
REGOUT
FSYNC
nCS
VDD_3V3
GND
GND
18
25
ICM-20649
GND

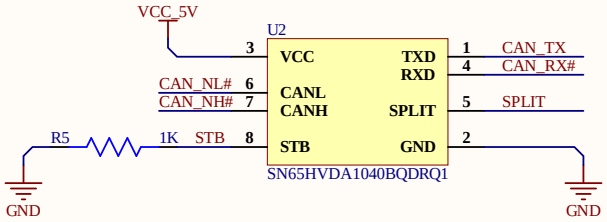

VDD_3V3
GND
VDD_3V3
GND

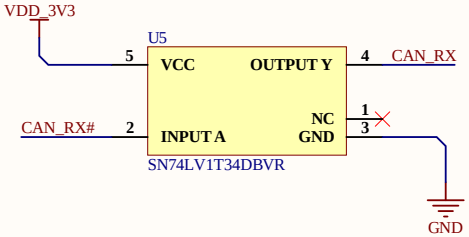


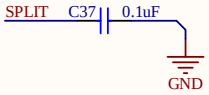


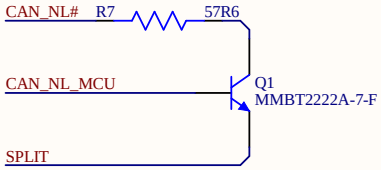


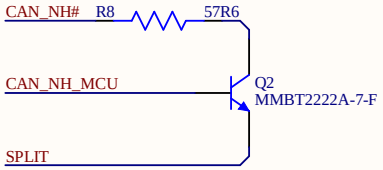
Connections for CAN are w.r.t Controller Rx and Tx

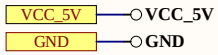


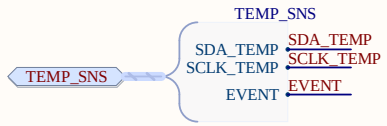


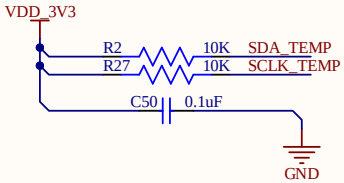


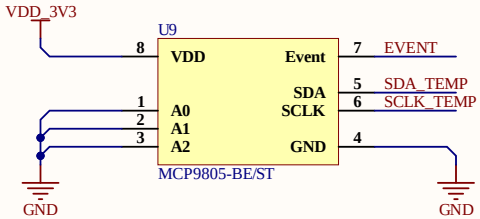


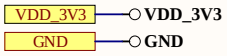


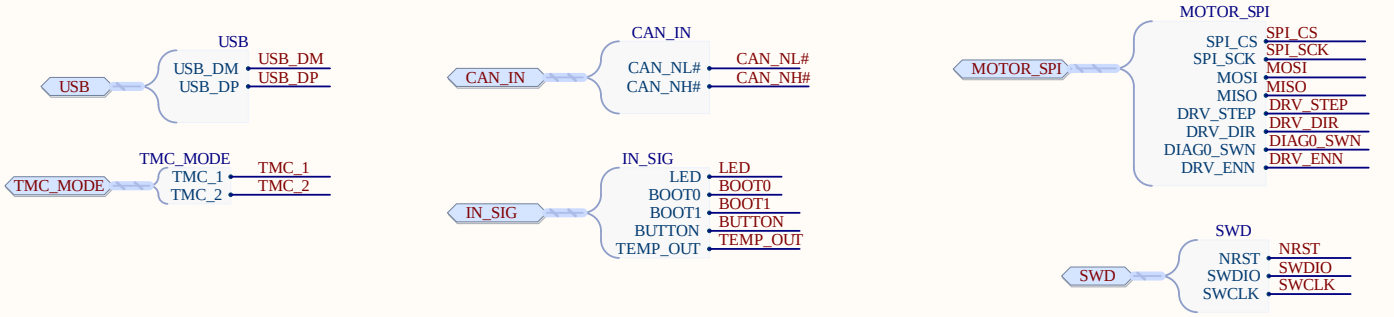


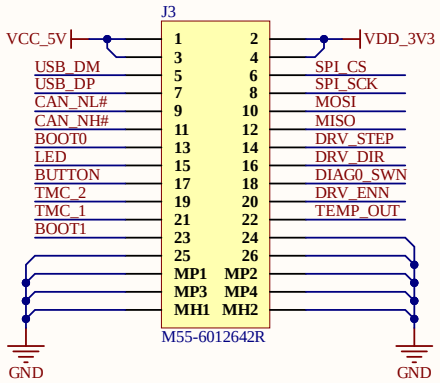

TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT

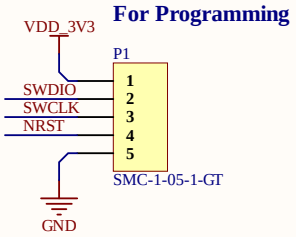

VDD_3V3
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND

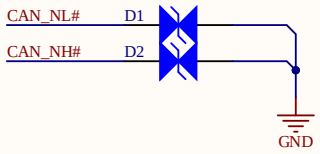

VDD_3V3
U9
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND

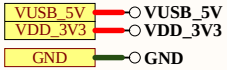

VDD_3V3
VDD_3V3
GND
GND


USB
USB_DM
USB_DP
USB_DM
USB_DP
CAN_IN
CAN_NL#
CAN_NH#
CAN_NL#
CAN_NH#
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
IN_SIG
LED
BOOT0
BOOT1
BUTTON
TEMP_OUT
LED
BOOT0
BOOT1
BUTTON
TEMP_OUT
SWD
NRST
SWDIO
SWCLK
NRST
SWDIO
SWCLK


J3
VCC_5V
VDD_3V3
USB_DM
USB_DP
CAN_NL#
CAN_NH#
BOOT0
LED
BUTTON
TMC_2
TMC_1
BOOT1
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TEMP_OUT
MP1
MP2
MP3
MP4
MH1
MH2
M55-6012642R
GND
GND


For Programming
VDD_3V3
P1
SWDIO
SWCLK
NRST
SMC-1-05-1-GT
GND


CAN_NL#
CAN_NH#
D1
D2
GND


VUSB_5V
VDD_3V3
GND
VUSB_5V
VDD_3V3
GND